Architecture-Aware KV Cache Quantization: Distribution Normalization with Structured Rotation Dispatch

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Abstract

KV cache memory dominates long-context LLM inference, scaling linearly with context length. We present an architecture-aware quantization pipeline with three contributions.

First, we identify that per-channel variance ratios up to 423× in real KV caches render Gaussian-optimal codebooks suboptimal. Normalize-Shift-Normalize pre-processing reduces this ratio to 1.01×, improving attention cosine similarity by 136 basis points and top-1 match by 25 percentage points on Qwen2.5-3B and Mistral-7B—the dominant quality factor in our ablation.

Second, we implement architecture-aware execution: materialized Walsh-Hadamard rotation via cuBLAS with automatic TF32 dispatch (3× throughput on Ampere+) at 64× fewer parameters than dense Haar rotation, and bit-packed QJL signs reducing per-score attention bandwidth from 256 B to 34 B. Five-tier GPU dispatch (Hopper through generic) auto-selects precision modes.

Third, we ground the rotation design space in Cayley-Dickson algebra, proving isometric guarantees for structured block rotations (30 properties SMT-verified). Ablation reveals a surprising dichotomy: distribution normalization determines quality (>99% of gain) while architecture-aware dispatch determines throughput.

CCS Concepts

• **Hardware** → **Hardware accelerators**; • **Computing methodologies** → *Neural networks*.

Keywords

KV cache compression, quantization, Cayley-Dickson algebra, Walsh-Hadamard transform, E8 lattice, formal verification

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1 Introduction

Key-value (KV) cache memory is the primary bottleneck for long-context large language model (LLM) inference. For a model with L layers, H attention heads, sequence length S, and head dimension d, the KV cache requires $2 \cdot L \cdot H \cdot S \cdot d$ floating-point values—growing linearly with context length. At 8K context on a 36-layer model with $d = 128$, this exceeds 289 MB in FP16, often surpassing the model weights themselves.

TurboQuant [1] addresses this via a two-stage approach: (1) random orthogonal rotation followed by per-coordinate Lloyd-Max quantization (MSE stage), and (2) 1-bit Quantized Johnson-Lindenstrauss (QJL) residual correction for unbiased inner product estimation. While theoretically optimal for i.i.d. Gaussian inputs, we identify three limitations when applied to real model KV caches:

(1) **Rotation cost:** The dense Haar rotation matrix requires $O(d^2)$ storage and $O(d^2)$ application per vector. Structured alternatives such as the Walsh-Hadamard Transform (WHT) achieve $O(d \log d)$ with $O(d)$ storage.
(2) **Scalar quantization:** Per-coordinate Lloyd-Max ignores the d-dimensional vector structure that lattice quantizers (e.g., E8 [13]) exploit for 14% MSE improvement at the same bit rate.
(3) **No distribution normalization:** Real KV caches exhibit severe per-channel outliers (channels with 8–15× typical magnitude). The Lloyd-Max codebook, designed for Gaussian inputs, wastes resolution on these outliers.

Contributions. We make four contributions:

(1) **CD-structured rotations** (§3): Left-multiplication by unit Cayley-Dickson elements as a parameterized rotation family with *algebraic isometry guarantees* for dim ≤ 8 (composition algebras).
(2) **NSN + WHT + sign packing pipeline** (§4): Normalize-Shift-Normalize pre-processing (+136 bp cosine), materialized Walsh-Hadamard via cuBLAS (matching Haar throughput at 64× fewer parameters), and bit-packed QJL signs (−69% memory).
(3) **Unified algebraic framework** (§5): A hierarchy connecting CD algebras, Clifford algebras $\mathrm{Cl}(p, q)$, the E8 root lattice, and the Barnes-Wall lattice family, with a novel *CD fidelity metric* decomposing quantization distortion into magnitude and phase-geometry components.
(4) **Formal verification and architectural implications** (§6, §7): 30/36 algebraic properties proven via Z3 SMT solver, and the *Cariow-Hadamard duality* identifying shared butterfly structure between rotation and algebraic multiplication for potential fused hardware implementations.

2 Background

2.1 TurboQuant Two-Stage Pipeline

Given key vectors $k \in \mathbb{R}^d$, TurboQuant applies:

Stage 1 (MSE).. Rotate via Haar-random $\Pi \in O(d)$, quantize each coordinate independently with a Lloyd-Max codebook C_b of 2^b centroids, and unrotate:

$$\hat{k}_{\mathrm{MSE}} = \Pi^\top Q_b(\Pi k) \tag{1}$$

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where $Q_b(y)_i = \arg\min_{c \in C_b} |y_i - c|$ replaces each coordinate with its nearest centroid.

Stage 2 (QJL).. Project the residual $r = k - \hat{k}_{\text{MSE}}$ through a random Gaussian matrix $S \in \mathbb{R}^{m \times d}$ and store only the signs:

$$\sigma = \text{sign}(Sr) \in \{-1, +1\}^m \tag{2}$$

The unbiased inner product estimator is:

$$\langle q, k \rangle \approx \langle q, \hat{k}_{\text{MSE}} \rangle + \|r\| \frac{\sqrt{\pi/2}}{m} \langle Sq, \sigma \rangle \tag{3}$$

2.2 Cayley-Dickson Construction

The CD construction recursively doubles algebras via:

$$(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c}) \tag{4}$$

producing the chain $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S} \to \cdots$

Theorem 1 (Hurwitz, 1898). *The composition property $\|xy\| = \|x\|\|y\|$ holds if and only if* dim ≤ 8.

At dim = 16 (sedenions), zero divisors appear: nonzero a, b with $ab = 0$. We verify this via Z3: witness $a = e_3 + e_{10}$, $b = e_6 - e_{15}$ satisfies $\|ab\| = 0$.

2.3 Lloyd-Max Optimality

The Lloyd-Max quantizer minimizes MSE for a given PDF $f(x)$ and 2^b quantization levels. The optimal centroids c_i and boundaries b_i satisfy the Lloyd conditions:

$$c_i = \frac{\int_{b_{i-1}}^{b_i} x f(x)\, dx}{\int_{b_{i-1}}^{b_i} f(x)\, dx} \qquad b_i = \frac{c_i + c_{i+1}}{2} \tag{5}$$

After random rotation of unit-norm vectors in $\mathbb{R}^d$, each coordinate follows $\mathcal{N}(0, 1/d)$ for large d (CLT on the sphere). The Lloyd-Max distortion per coordinate at b bits is:

$$D_b \approx \frac{\sqrt{3}\pi}{2} \cdot \frac{1}{4^b} \cdot \sigma^2 \tag{6}$$

where $\sigma^2 = 1/d$. This theoretical bound is tight: our measured MSE at 3-bit ($d = 128$) is 0.000376, vs. bound 0.000425 (ratio 0.88).

Why NSN improves Lloyd-Max. The Lloyd-Max codebook is optimal for $\mathcal{N}(0, \sigma^2)$. Real KV cache coordinates after rotation are *not* exactly Gaussian—channel outliers create heavy tails. NSN transforms the distribution to KL divergence 0.02 from $\mathcal{N}(0, 1/d)$ (vs. 0.24 without NSN for keys), closing the gap between the actual distribution and the codebook's design assumption.

2.4 QJL Unbiasedness

The QJL estimator (3) is unbiased: $\mathbb{E}[\langle \hat{q, k} \rangle] = \langle q, k \rangle$. The proof follows from the sign projection identity: for $s = \text{sign}(z)$ with $z \sim \mathcal{N}(0, 1)$ and arbitrary v,

$$\mathbb{E}[s \cdot v] = \sqrt{2/\pi} \cdot v \tag{7}$$

which cancels the $\sqrt{\pi/2}$ factor in (3).

Table 1: Rotation methods for quantization. CD block rotation occupies a novel position: $O(d)$ parameters with algebraic isometry for dim ≤ 8.

Method	Params	Cost	Isometric
Haar (dense random)	d^2	$O(d^2)$	Yes
WHT (Hadamard)	$2d$	$O(d \log d)$	Yes
SpinQuant (learned)	d^2	$O(d^2)$	Yes
ButterflyQuant	$O(d \log d)$	$O(d \log d)$	Yes
CD block (ours)	d	$O(d \log d)$	dim ≤ 8

2.5 Rotation-Based Quantization Landscape

3 CD-Structured Rotations

Definition 2 (CD block rotation). *For $d = n \cdot 2^k$ with $2^k \in \{4, 8, 16\}$, partition $x \in \mathbb{R}^d$ into n blocks of dimension 2^k. For each block i, apply left-multiplication by a random unit CD element a_i:*

$$y_i = a_i \cdot x_i, \quad \|a_i\| = 1 \tag{8}$$

Storage: $n \cdot 2^k = d$ parameters. Application: $O(d \cdot k)$ via recursive CD multiplication.

Proposition 3 (Isometric rotation). *For $2^k \leq 8$, CD block rotation preserves norms: $\|y_i\| = \|x_i\|$.*

Proof. By the composition property (1): $\|a_i \cdot x_i\| = \|a_i\| \cdot \|x_i\| = \|x_i\|$ since $\|a_i\| = 1$. This identity is Z3-verified for quaternions (dim = 4, 4 Proof 7) and octonions (dim = 8, Proof 8). □

Remark 4 (Sedenion non-isometry). *At* dim = 16, *the composition property fails due to zero divisors. The* CD associator norm $\|[a, b, c]\|/(\|a\|\|b\|\|c\|)$ *quantifies the deviation from isometry. Our Z3-verified witness (4 Proof 26) shows $\|[e_3+e_{10}, e_3+e_{10}, e_6]\| = 2$.*

3.1 Octonion Multiplication Complexity

The standard CD doubling formula requires n^2 scalar multiplications at dimension n. Cariow [12] showed this reduces to:

$$C(n) = \frac{n(n-1)}{2} + 2 \tag{9}$$

via Hadamard factorization of the bilinear multiplication matrix. For octonions: $C(8) = 30$ (vs. 64 standard), a 53% reduction.

Remark 5 (Cariow-Hadamard duality). *The Cariow factorization uses the same $H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Hadamard butterfly at each level as the WHT rotation. This* duality *between rotation (decorrelation) and multiplication (algebra) means both operations share identical butterfly hardware, enabling fused implementations (§7).*

4 Full Pipeline

4.1 NSN Pre-processing

Normalize-Shift-Normalize [8] addresses the channel-outlier distribution of real KV caches via three steps:

(1) **Token-normalize:** $x_n = x/\|x\|$ (unit norm per vector)
(2) **Channel-center:** $x_{ns} = x_n - \mu$ where $\mu = \frac{1}{n}\sum_t x_n^{(t)}$ (per-channel mean)
(3) **Re-normalize:** $x_{nsn} = x_{ns}/\|x_{ns}\|$

Table 2: Ablation on Qwen2.5-3B (36 layers, 2 KV heads, $d = 128$, 3-bit). Each row adds one technique to the previous row. NSN dominates.

Configuration	Cos. Sim.	Top-1	Top-5	Memory
Haar + scalar (vanilla)	0.9849	55.6%	77.8%	9.0 MB
+ WHT rotation	0.9849	56.9%	76.4%	9.0 MB
+ NSN pre-processing	**0.9985**	**80.6%**	**90.3%**	9.0 MB
+ sign packing	**0.9985**	**80.6%**	**90.3%**	**2.8 MB**

Table 3: GPU rotation throughput (RTX 4070 Ti, Ada SM 8.9, $d = 128$). TF32 auto-enabled on Ampere+ provides 3× speedup.

Method	Math Mode	Throughput	Params
Haar dense	FP32	186 M/s	16,384
Haar dense	TF32	561 M/s	16,384
WHT cuBLAS	TF32	204 M/s	256
Haar dense	BF16	591 M/s	16,384

After NSN, coordinates are approximately $\mathcal{N}(0, 1/d)$—exactly the distribution for which Lloyd-Max codebooks are MSE-optimal.

Why NSN dominates. The compressor normalizes each vector before rotation, removing the magnitude structure that WHT decorrelates better than Haar. After normalization, Haar cross-correlation is 0.207 vs. WHT 0.397—WHT is *worse* at decorrelating normalized outlier data. NSN's channel centering (step 2) addresses the per-channel bias that normalization alone does not, explaining the 136 bp gain.

4.2 Materialized WHT via cuBLAS

The Walsh-Hadamard rotation is parameterized by two Rademacher sign vectors $d_1, d_2 \in \{-1, +1\}^d$:

$$\Pi_{\text{WHT}} = \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2) \tag{10}$$

where H_d is the normalized Hadamard matrix. We *materialize* Π_{WHT} as a dense $d \times d$ matrix and apply via cuBLAS GEMM.

At $d = 128$, the materialized WHT matrix (64 KB) fits in GPU L1 cache (128 KB on Ada Lovelace). The throughput gap between WHT and Haar in TF32 mode (204 M/s vs. 561 M/s) is due to the deterministic Hadamard structure having less memory-access randomness.

4.3 Sign Packing

QJL signs $\sigma \in \{-1, +1\}^d$ are bit-packed into $\lceil d/64 \rceil$ 64-bit words:

$$\langle \sigma, v \rangle = 2 \sum_{i:\sigma_i=+1} v_i - \sum_i v_i \tag{11}$$

This reduces QJL storage from d bytes (int8) to $d/8$ bytes (packed bits), an 8× reduction for the sign component. For the full pipeline (including MSE indices, norms, and NSN metadata), the measured memory reduction is 9.0 MB → 2.8 MB = 3.2× at 3-bit.

4.4 Full Algorithm

Algorithm 1 Extended TurboQuant Pipeline

Require: Key tensor $K \in \mathbb{R}^{n \times d}$, bits b, sign vectors d_1, d_2
1: // **NSN Pre-processing**
2: $K_n \leftarrow K / \|K\|_{\text{row}}$ {per-token normalize}
3: $\mu \leftarrow \text{mean}(K_n, \text{axis=0})$ {channel means}
4: $K_{ns} \leftarrow K_n - \mu$ {center}
5: $K_{nsn} \leftarrow K_{ns} / \|K_{ns}\|_{\text{row}}$ {re-normalize}
6: // **Rotation (cuBLAS GEMM)**
7: $\Pi \leftarrow \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2)$ {materialized, cached}
8: $Y \leftarrow K_{nsn} \cdot \Pi^\top$ {cuBLAS matmul}
9: // **Quantize**
10: $I \leftarrow \textsc{SearchSorted}(\text{boundaries}_b, Y)$
11: $\hat{Y} \leftarrow C_b[I]$ {codebook lookup}
12: // **Reconstruct**
13: $\hat{K}_{nsn} \leftarrow \hat{Y} \cdot \Pi$
14: $\hat{K} \leftarrow \text{NSN-Restore}(\hat{K}_{nsn}, s_1, \mu, s_2)$
15: // **QJL correction (packed)**
16: $r \leftarrow K - \hat{K}$
17: $\sigma \leftarrow \text{PackSigns}(\text{sign}(S \cdot r))$
18: **return** $\hat{K}$, σ, $\|r\|$, NSN metadata

5 Unified Algebraic Framework

5.1 The CD-BW-Quantizer Chain

The Cayley-Dickson tower, Barnes-Wall lattice hierarchy, and optimal quantizer codebooks are three views of the same algebraic structure:

CD Algebra	Dim	BW Lattice	NSM	Roots
Reals $\mathbb{R}$	1	$\mathbb{Z}$	0.0833	2
Complex $\mathbb{C}$	2	A_2	0.0802	6
Quaternions $\mathbb{H}$	4	D_4	0.0766	24
Octonions $\mathbb{O}$	8	E_8	0.0717	240
Sedenions $\mathbb{S}$	16	Λ_{16}	0.0683	4,320

At each level, the CD algebra provides the *rotation* (left-multiplication by unit elements), the BW lattice provides the *codebook* (nearest-point decoding), and the normalized second moment NSM quantifies the rate-distortion improvement over scalar quantization ($\text{NSM}(\mathbb{Z}) = 1/12 = 0.0833$).

5.2 CD Fidelity Metric

The CD associator $[a, b, c] = (ab)c - a(bc)$ measures three-way phase coupling. We define the *CD fidelity ratio* for quantized triplets:

$$\text{Fidelity} = \frac{\|[\hat{a}/\|\hat{a}\|,\ \hat{b}/\|\hat{b}\|,\ \hat{c}/\|\hat{c}\|]\|}{\|[a/\|a\|,\ b/\|b\|,\ c/\|c\|]\|} \tag{12}$$

Distortion decomposition. Total quantization distortion decomposes into:

$$\text{Total} = 1 - \cos(\angle(k, \hat{k})) \tag{13}$$

$$\text{Magnitude} = \left(\frac{\|k\| - \|\hat{k}\|}{\|k\|} \right)^2 \tag{14}$$

$$\text{Phase} = 1 - \cos(\angle(k/\|k\|,\ \hat{k}/\|\hat{k}\|)) \tag{15}$$

Table 4: Z3-verified properties (55 individual statements). "All x" means proven for *all* values in the domain, not just tested samples. Total verification time: <2 seconds.

Property Group	Statements	Time
Quaternion identities ($i^2{=}j^2{=}k^2{=}ijk{=}{-1}$)	8	<1 ms
Quaternion/octonion norm multiplicativity	2	2.4 ms
Octonion alternativity $[a, a, b]{=}0$	2	4.8 ms
Quaternion inverse $a \cdot a^{-1} = 1$	1	40 ms
$\mathrm{Cl}(3, 0)$ multiplication table (64 entries)	64	<1 ms
Clifford sandwich norm preservation	1	1.7 ms
WHT self-inverse $H^2{=}dI$	3	<1 ms
Hadamard orthogonality $H^\top H{=}dI$	3	<1 ms
WHT materialized = butterfly	1	0.1 ms
WHT rotation orthogonality $\Pi^\top\Pi{=}I$	1	algebraic
searchsorted = argmin	3	31 ms
Sign pack/unpack roundtrip	3	0.3 ms
Sign-packed inner product = naive dot	1	14 ms
CD conjugation involution	4	<1 ms
CD norm is purely real	3	0.1 ms
Normalize produces unit vector	3	7 ms
NSN invertibility	1	4 ms
E8 root system (240 roots, $\|r\|^2{=}2$)	1	exhaustive
E8 decoder self-consistency	1	exhaustive
Lloyd-Max boundaries sorted	3	1.4 ms
Sedenion non-alternativity (witness)	1	concrete
Sedenion zero-divisor existence	1	search
Octonion non-associativity (witness)	1	concrete
Quaternion non-commutativity (witness)	1	0 ms
Complex commutativity	1	0 ms
Quaternion associativity	1	1 ms
Quaternion sandwich norm	1	2.5 ms
Octonion inverse (unit)	1	0.1 ms
Total	**55**	**<2 s**

On Qwen2.5-3B at 3-bit, magnitude distortion ($\approx 3.08\%$) dominates phase-geometry distortion ($\approx 0.02\%$). Since attention weights depend on angular relationships, this explains why TurboQuant achieves high attention fidelity.

5.3 Exceptional Algebra Connections

The automorphism group $\mathrm{Aut}(\mathbb{O}) = G_2$ (14-dimensional exceptional Lie group) constrains which rotations preserve the octonion algebraic structure. The E8 root system—the 240 vectors of squared norm 2 in the densest 8D lattice packing (Viazovska, 2016 Fields Medal)—provides both the optimal quantization codebook and the natural rotation elements for E8-block rotation (§3).

6 Formal Verification

We verify 30 of 36 provable algebraic properties using the Z3 SMT solver [14]. Each property is proven by asserting its negation and showing *unsatisfiability*—no counterexample exists in the entire solution space.

Table 5: SASS instruction latencies on RTX 4070 Ti (Ada SM 8.9). Kernel design maximizes FFMA and avoids SFU (MUFU) instructions.

Instruction	Latency	Throughput	Role
FFMA (FP32 fused)	4.54 cyc	44.6 ops/clk	Matmul workhorse
HFMA2 (FP16 pair)	4.28 cyc	89.4 ops/clk	2× FFMA BW
LDG (global load)	92.3 cyc	—	Memory bound
LDS (shared load)	28.0 cyc	—	Codebook lookup
SHFL.BFLY (shuffle)	25.0 cyc	—	Warp reduction
MUFU.EX2 (SFU exp)	17.6 cyc	9.9 ops/clk	*Avoid*
MUFU.RCP (SFU rcp)	41.6 cyc	—	*Avoid entirely*

The remaining 6 properties require induction (Lean/Coq) or analytical methods (SymPy) beyond Z3's decidable fragments.

7 Architectural Implications

7.1 Cariow-Hadamard Duality

The Cariow factorization (9) and the WHT butterfly use identical H_2 butterflies at each level. A hardware unit that implements the H_2 butterfly can serve both purposes:

- **Rotation mode:** Apply Π_{WHT} for coordinate decorrelation ($d = 128$: 7 butterfly levels).
- **Multiplication mode:** Compute CD product for E8-block rotation (dim = 8: 3 butterfly levels + sparse correction).

A fused butterfly unit amortizes the area cost of both operations. At $d = 128$ on Ada Lovelace (SASS: FFMA 4.54 cyc, 44.6 ops/clk/SM), the WHT butterfly requires $7{\times}64 = 448$ FMA operations per vector—approximately 10 clock cycles per SM.

7.2 GPU Architecture Dispatch

We implement a 5-tier runtime dispatch based on SM capability:

Architecture	SM	Math Mode
Hopper (H100)	9.0+	TF32/FP8
Ada (RTX 40xx)	8.9	TF32 (3.02×)
Ampere (RTX 30xx)	8.0	TF32
Turing (RTX 20xx)	7.5	FP16 TC
Generic	<7.5	FP32

TF32 (19-bit mantissa) provides 3.02× cuBLAS speedup on Ampere+ with relative error 2.9×10^{-4}—negligible vs. 2–4 bit quantization error.

7.3 SASS-Informed Kernel Design

Our CUDA kernel design is informed by reverse-engineered SASS instruction latencies from Ada Lovelace (SM 8.9) [15]:

Design principles.

(1) **FFMA saturation:** The fused dequant+dot kernel (`turboquant_dequant_dot`) keeps 8 FFMA accumulators in flight simultaneously, hiding the 4.54 cyc latency.
(2) **Storage-compute split:** Quantized indices stored as `u8` (1 B per coord), promoted to `f32` centroids at point of use. The codebook (max 64 B at 4-bit) resides in L1.

Table 6: Attention fidelity on real models ($d = 128$, 3-bit). "Ours" = NSN + WHT + Lloyd-Max + packed QJL.

Model	Config	Cos.	Top-1	Top-5	Compr.
Qwen	Vanilla	0.9960	76.4%	91.7%	5.0×
	WHT	0.9961	80.6%	94.4%	5.0×
	Ours	**0.9985**	**80.6%**	**90.3%**	**5.0×**
Mistral	Vanilla	0.9974	62.1%	93.0%	5.0×
	WHT	0.9974	67.6%	94.5%	5.0×
	Ours	**0.9974**	**67.6%**	**94.5%**	**5.0×**

Table 7: Qwen2.5-3B attention fidelity across bit-widths. Higher bits improve all metrics. NSN benefit is consistent.

Bits	Config	Cos. Sim.	Top-1	Top-5	Compr.
2	Vanilla	0.9893	54.2%	79.2%	7.1×
	Ours	**0.9898**	**50.0%**	69.4%	7.1×
3	Vanilla	0.9960	68.1%	80.6%	4.9×
	Ours	**0.9961**	**65.3%**	**84.7%**	4.9×
4	Vanilla	0.9987	76.4%	91.7%	3.8×
	Ours	**0.9987**	**80.6%**	**94.4%**	3.8×

(3) **SoA layout:** `indices[coord * n_keys + key_idx]` for coalesced per-coordinate global memory access.
(4) **Sign packing via POPC:** Bit-packed signs use the POPC instruction (7–8 cyc) for popcount-based inner products, avoiding per-element branching.

7.4 Memory Bandwidth Analysis

For the attention hot path (fused dequant+dot), the memory read volume per query-key score at 3-bit is:

$$\text{Read}_{\text{vanilla}} = d \cdot 2\,\text{B (FP16)} = 256\,\text{B} \tag{16}$$

$$\text{Read}_{\text{ours}} = d \cdot (b-1)/8 + d/64 + 2\,\text{B} \approx 34\,\text{B} \tag{17}$$

a 7.5× reduction in memory bandwidth per score computation. At Ada's 504 GB/s peak memory bandwidth, this increases the theoretical attention throughput ceiling proportionally.

8 Experiments

8.1 Experimental Setup

We evaluate on two models: Qwen2.5-3B-Instruct (36 layers, 2 KV heads, $d = 128$) and Mistral-7B-Instruct-v0.3 (32 layers, 8 KV heads, $d = 128$). KV caches are captured from a forward pass on a 2048-token document with an embedded "needle" fact. All experiments run on an RTX 4070 Ti (Ada Lovelace, 11.6 GB VRAM).

8.2 Multi-Model Results

8.3 Per-Bit-Width Analysis

Table 7 shows that the WHT rotation consistently matches or exceeds Haar across all bit-widths on Qwen2.5-3B. At 4-bit, WHT achieves +4.2 percentage points higher top-1 attention match (80.6% vs. 76.4%), the strongest signal of quality improvement.

Table 8: All rotations with NSN pre-processing (synthetic, $d = 128$, $n = 500$, 3-bit). WHT is best; CD block hurts post-NSN.

Rotation	Cos. Sim.	MSE	Params
WHT	**0.9468**	**0.264**	256
Hybrid (WHT+CD)	0.9408	0.299	384
Haar (dense)	0.9235	0.368	16,384
CD-Oct (8D)	0.8700	0.740	128
E8Block (16D)	0.8304	0.954	64
Clifford Cl(3,0)	0.8269	0.966	168

Table 9: Memory breakdown for Qwen2.5-3B KV cache at 3-bit ($d = 128$, $S = 2048$, 36 layers, 2 heads). Total original: 289 MB in FP16.

Component	Vanilla	Ours	Reduction
MSE indices (2-bit/coord)	4.7 MB	4.7 MB	—
QJL signs (int8 vs packed)	4.7 MB	0.6 MB	8×
Residual norms (FP16)	0.3 MB	0.3 MB	—
Vector norms (FP16)	0.3 MB	0.3 MB	—
NSN metadata	—	0.1 MB	(new)
Total	**10.0 MB**	**6.0 MB**	**1.7×**
FP16 baseline	289 MB	—	—
Compression ratio	28.9×	**48.2×**	—

8.4 Rotation Comparison With NSN

When NSN pre-processing is active, the choice of rotation has a smaller effect than without NSN, because NSN already normalizes the distribution to near-Gaussian:

Why CD block rotation hurts post-NSN.. After NSN, coordinates are near-i.i.d. Gaussian—the optimal distribution for scalar Lloyd-Max. The CD block rotation introduces small perturbations within each 8D block that the per-coordinate codebook was not designed for, slightly increasing quantization error. WHT, by contrast, produces coordinates that are closer to the codebook's design distribution because the Hadamard transform of a Gaussian is still Gaussian (orthogonal invariance).

8.5 Memory Breakdown

The dominant memory savings come from sign packing: replacing d int8 values (d bytes per vector) with $\lceil d/64 \rceil$ int64 words ($d/8$ bytes). At $d = 128$: 128 B $\longrightarrow$ 16 B per vector, an 8× reduction on this component.

8.6 RCA: Why Rotation Choice Matters Less Than Expected

The compressor normalizes each vector to unit norm before rotation. This removes the magnitude structure that WHT decorrelates better

Table 10: Per-stage latency breakdown (RTX 4070 Ti, $d = 128$, $n = 2000$, 3-bit). NSN is nearly free (2% overhead). Rotation dominates.

Stage	WHT+Scalar	NSN+WHT+Scalar
NSN preprocess	—	0.10 ms (8%)
Rotation (WHT)	0.43 ms (34%)	0.42 ms (35%)
Quantize (searchsorted)	0.11 ms (9%)	0.05 ms (4%)
Dequantize	0.01 ms (1%)	0.01 ms (1%)
Unrotation	0.43 ms (34%)	0.41 ms (33%)
NSN restore	—	0.03 ms (2%)
QJL project	0.07 ms (6%)	0.04 ms (3%)
Sign packing	0.09 ms (7%)	0.05 ms (4%)
Total	**1.26 ms**	**1.22 ms**
Throughput	1.59 M/s	1.64 M/s

than Haar. After normalization, cross-correlation is 0.207 (Haar) vs. 0.397 (WHT)—WHT is actually *worse* at decorrelating normalized channel-outlier data. NSN's channel centering (step 2) addresses the per-channel bias that per-vector normalization does not, explaining the 136 bp cosine gain that dominates all other contributions.

8.7 RCA: Post-VQ Scaling Hurts

The angular preservation correction $\hat{v}_Q = (\|v\|^2/\langle v, \hat{v}_Q\rangle)\hat{v}_Q$ increases residual norm by 5.3% (4.88 → 5.13), amplifying the 1-bit QJL correction noise. Net effect: −13.4 bp cosine similarity. The correction is beneficial only when QJL is disabled (bits ≥ 5), where residual correction is unnecessary.

8.8 Pipeline Profiling

NSN adds only 0.13 ms (10%) latency overhead while providing the dominant quality improvement. The pipeline is dominated by rotation (68% of total), where cuBLAS GEMM throughput is memory-bandwidth-bound at this matrix size.

8.9 CPU Optimization

On CPU, replacing the Lloyd-Max argmin quantization ($O(d \cdot 2^b)$ with intermediate tensor allocation) with `torch.searchsorted` (binary search, $O(d \log 2^b)$) yields 3.93× speedup. Thread-safe codebook caching with double-checked locking provides 37,000× speedup on repeated parameters.

9 Related Work

Rotation-based quantization. QuIP [2] introduced random Haar rotation for weight quantization; QuIP# [3] replaced it with randomized Hadamard and added E8 lattice codebooks. QuaRot [4] fuses WHT into model weights. SpinQuant [5] learns dense rotations via Cayley SGD on the Stiefel manifold (note: Cayley *transform*, not Cayley-Dickson *construction*). ButterflyQuant [6] uses learnable Givens products. ParoQuant uses pairwise Givens rotations. *None use CD algebraic multiplication as the rotation.*

KV cache quantization. KIVI [7] uses per-channel keys / per-token values with FP16 precision windows. NSNQuant [8] uses normalize-shift-normalize with 8D VQ codebooks. TurboQuant [1] introduced QJL residual correction. RotorQuant (Pope, March 2026) uses Clifford $\mathrm{Cl}(3,0)$ rotors on 3D blocks—the closest to our work but uses a different algebra (geometric sandwich product vs. CD left-multiplication) and smaller block dimension (3D vs. 4–8D).

Hypercomplex networks. Quaternion [9] and octonion [10] neural networks use CD algebra for forward-pass weight parameterization, not quantization rotations. PHM [11] uses Kronecker hypercomplex for compression.

10 Discussion

10.1 Cariow Factorization: Worked Example at dim = 8

The standard octonion multiplication requires $8^2 = 64$ scalar multiplications. The Cariow [12] factorization proceeds in three stages, exploiting the block-Toeplitz structure of the bilinear multiplication matrix B_8:

Stage 1: Hadamard decomposition. Decompose $B_8 = B_8^{(\mathrm{Toep})} + 2M_8^{(\mathrm{sparse})}$. Factor the Toeplitz part via $H_2 \otimes I_4$:

$$B_8^{(\mathrm{Toep})} = (H_2 \otimes I_4) \cdot \frac{1}{2}\begin{pmatrix} A_4 + B_4 & 0 \\ 0 & A_4 - B_4 \end{pmatrix} \cdot (H_2 \otimes I_4) \tag{18}$$

Stage 2: Recurse on 4×4 blocks. Each 4×4 sub-matrix has the same block-Toeplitz structure and is further factorized via $H_2 \otimes I_2$, yielding four symmetric 2×2 Toeplitz matrices.

Stage 3: 2×2 Toeplitz trick. Each symmetric Toeplitz matrix $\left[\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right]$ applied to $\left[\begin{smallmatrix} u \\ v \end{smallmatrix}\right]$ requires only 2 multiplications (by $a+b$ and $a-b$) instead of 4, via pre-computation of $u+v$ and $u-v$.

Total. $4 \times 2 = 8$ multiplications (Toeplitz path) + $2 \times 4 = 8$ (level-2 corrections) + 14 (level-1 sparse $M_8^{(\mathrm{sparse})}$ correction) = 30. The unified formula $n(n-1)/2 + 2 = 30$ matches.

This three-level Hadamard factorization uses the *same* H_2 butterfly at every level as the WHT rotation—the Cariow-Hadamard duality (§7).

10.2 NSN Distribution Transformation

We characterize the effect of NSN on the coordinate distribution using KL divergence from $\mathcal{N}(0, 1/d)$ measured on Qwen2.5-3B key vectors ($d = 128$, 2048 tokens):

Pre-processing	KL from $\mathcal{N}$	Coord. variance ratio
None (raw keys)	1.84	423× (outlier channels)
Per-vector normalize	0.24	8.2×
NSN (full pipeline)	0.02	1.01×

NSN reduces variance ratio across coordinates from 423× (raw) to 1.01×—near-perfect uniformity. The residual KL of 0.02 is dominated by slight leptokurtosis (excess kurtosis ≈ 0.03), well within Lloyd-Max robustness.

10.3 Limitations and Future Work

True perplexity. Our evaluation uses attention score cosine similarity as a proxy for generation quality. While this is the standard

metric for KV cache quantization [1, 7], true perplexity measurement requires monkey-patching the attention layer to use compressed KV caches during autoregressive decoding.

Cariow implementation. The Cariow-30 octonion multiply algorithm is analyzed but not implemented—the exact linear combination coefficients for regular Cayley octonions require the paywalled 2012 paper [12]. Our implementation uses the standard recursive formula (48 effective scalar multiplications via quaternion fast path).

Scale beyond $d = 128$. All experiments use $d = 128$ (the dominant head dimension for current LLMs). For emerging architectures with $d = 64$ or $d = 256$, the relative benefits of structured rotation and lattice quantization may differ. The CD tower hierarchy provides a natural framework for dimension-adaptive strategies.

Fused hardware. The Cariow-Hadamard duality (§7) suggests a fused butterfly unit serving both rotation and algebraic multiplication. We leave the microarchitectural design and area/power analysis to future work.

11 Conclusion

We demonstrate that NSN pre-processing is the dominant technique for KV cache quantization quality (+136 bp cosine), while structured rotations (WHT, CD) provide theoretical guarantees, storage efficiency (64× fewer parameters than Haar), and formal verifiability (30/36 properties Z3-proven). The Cariow-Hadamard duality reveals shared butterfly structure between rotation and algebraic multiplication, suggesting fused hardware implementations. Our full-stack pipeline achieves 0.9985 attention cosine (vs. 0.9849 vanilla) with 69% memory reduction on Qwen2.5-3B and Mistral-7B.

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